Invention Disclosure Format (IDF)-B - completed research response

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Prepared 20 October 2026. Expanded layout preserves the blank form's ten questions, order and numbering. The original blank is unchanged. This response supersedes the earlier ScopeGuard annex's unresolved novelty framing. No defensible novelty has been established; this is a research-paper-only recommendation, not an assertion of patentability. Investigator identity, ownership and signatures are not invented.

1. Title of the invention:

ClaimAudit / ScopeGuard: Reusable Source-Interpretation Verification for Numerical Scientific Claims (research prototype using established methods).

2. Field /Area of invention:

Computer-assisted scientific claim verification; uncertain relational data; knowledge compilation; numerical predicate verification; reproducible evaluation of context-consistent evidence. Intended use: review a compound numerical assertion against a finite set of scientific table configurations and explicitly represented alternative source interpretations. Financial fraud is outside this project.

3. Prior Patents and Publications from literature (provide a table summarizing the prior art)

The following sources were opened during Stage 1 on 19 October 2026. Dates are publication/submission dates where verified. A missing feature means it was not established in the inspected material, not that the entire source legally excludes it. Abstract-only and date limitations are stated. The full search-query record and excluded URLs are in STAGE1_PRIOR_ART.md.

Source / date / URL	Coverage and remaining limits
P01 - Probabilistic Databases: Diamonds in the Dirt (Extended Version) Date: 2009 bibliographic year; exact publication date was not verified from the opened author PDF. URL: https://www-cs.stanford.edu/~chrismre/papers/cacm-paper-full.pdf	Covers probabilistic databases, BID/x-tuple alternatives, correlated tuples, provenance and possible-world query evaluation. Does not establish an implemented scientific-table extraction workflow. That application difference does not make the representation new.
P02 - Consensus Answers for Queries over Probabilistic Databases Date: 10 December 2008 arXiv submission; PODS 2009 publication. URL: https://arxiv.org/abs/0812.2049	Covers richer and/xor representations beyond x-tuples/BID and answer selection over probabilistic worlds. Does not present ScopeGuard's specific four-status interface. Inspected abstract and author paper search record; no claim-by-claim theorem equivalence established.
P03 - Certain Answers as Objects and Knowledge Date: 2016 bibliographic year; author PDF verified, exact publication date not independently verified here. URL: https://homepages.inf.ed.ac.uk/libkin/papers/aij.pdf	Covers certainty under incomplete information and alternative answer representations. Does not supply our scientific-table ingestion or fixed predicate grammar. Universal reasoning over admissible interpretations is established, not a ScopeGuard invention.
P04 - Graph-Based Algorithms for Boolean Function Manipulation Date: August 1986, explicitly stated on page 1 of the author's updated copy. URL: https://www.cs.cmu.edu/~bryant/pubdir/ieeetc86.pdf	Covers canonical ordered decision graphs and operations sharing equivalent Boolean structure, with exponential worst cases. Does not specialize its representation to scientific claims. Renaming equivalent-state compression or cofactoring is not a technical distinction.
P05 - New Limits for Knowledge Compilation and Applications to Exact Model Counting Date: 8 June 2015; revised 19 August 2015. URL: https://arxiv.org/abs/1506.02639	Covers representation-size limitations, including query problems in probabilistic databases. Does not establish a universal efficient solution; it warns against deriving general scalability from favorable examples. Abstract inspected.
P06 - Sensitivity Analysis and Explanations for Robust Query Evaluation in Probabilistic Databases Date: SIGMOD 2011, pages 841-852, verified on the research publication page. URL: https://research.google/pubs/sensitivity-analysis-and-explanations-for-robust-query-evaluation-in-probabilistic-databases/	Covers influential inputs, explanations and incremental evaluation when uncertainty changes. Does not specifically describe a scientific header UI. Recomputing only relevant uncertainty is not defensibly new on that basis.
P07 - Query-Guided Resolution in Uncertain Databases Date: 2023 bibliographic year, DOI 10.1145/3589325; exact date not verified because publisher access failed. Author full text verified. URL: https://u.cs.biu.ac.il/~amstery/files/drien2023query.pdf	Covers Boolean provenance and selection of tuples for an oracle to resolve until query answers are determined. Does not establish the exact cheapest-single-header rule of ScopeGuard. That restriction is a straightforward special case, not evidence of inventive step.
P08 - Cleaning Uncertain Data with Quality Guarantees Date: August 2008, PVLDB 1(1), pages 722-735, printed in the PDF. URL: https://www.vldb.org/pvldb/vol1/1453935.pdf	Covers possible-world answer quality and resource-constrained selection of objects to clean. Does not concern language-model scientific narratives. Budget and query-result-guided clarification are already central mechanisms.
P09 - RefChecker: Reference-based Fine-grained Hallucination Checker and Benchmark for Large Language Models Date: 23 May 2024. URLs: https://arxiv.org/abs/2405.14486 ; https://arxiv.org/html/2405.14486v1	Covers claim-triplet extraction and checking against references. Does not establish exhaustive correlated source-world semantics. Claim decomposition or a verifier pipeline must not be claimed new.

Source / date / URL	Coverage and remaining limits
P10 - Proof-Carrying Numbers (PCN): A Protocol for Trustworthy Numeric Answers from LLMs via Claim Verification Date: 8 September 2025. URLs: https://arxiv.org/abs/2509.06902 ; https://arxiv.org/html/2509.06902v1	Covers source-referenced numerical checks under declared policies and verified presentation. Does not establish joint existential context satisfaction across every alternative source binding. Numeric provenance, deterministic checks and verification records are already known.
P11 - AmbiFC: Fact-Checking Ambiguous Claims with Evidence Date: first submitted 1 April 2021; inspected current abstract revised 14 December 2023. URL: https://arxiv.org/abs/2104.00640	Covers ambiguity and conflicting valid claim/evidence interpretations. Does not establish the present exact numerical context engine in the inspected abstract. Ambiguity reporting itself is not novel.
P12 - Same Claim, Different Interpretations: How LLM Agents Resolve Ambiguity in Scientific Claims Date: 14 October 2026. URLs: https://arxiv.org/abs/2610.20457 ; https://arxiv.org/html/2610.20457v1	Covers explicit Claim Cards, provenance of interpretation fields, interventions and changed verdicts/tool behavior. Does not establish enumeration of correlated table-header alternatives or the present single-header minimax rule. Explicit interpretations and their influence on verdicts cannot be claimed new.
P13 - When Verification Fails: How Compositionally Infeasible Claims Escape Rejection Date: 13 April 2026. URL: https://arxiv.org/abs/2604.10990	Covers failures on compound scientific constraints and salient-constraint shortcuts. Does not establish ScopeGuard's exact execution engine in the inspected abstract. The failure mode "individual pieces look supported but the whole claim fails" is not new.
P14 - US20260127386A1: Language model theory solvers Date: published 7 May 2026; listed priority 1 November 2024 (priority not legally verified). URL: https://patents.google.com/patent/US20260127386A1/en	Claims 1-3 concern LLM-derived constraints, jointly satisfiable values and SMT-assisted processing. They do not expressly recite our whole correlated-header protocol in those inspected claims. Generic solver-assisted joint constraint verification is not a distinguishing feature. No infringement conclusion follows.
P15 - US20250077940A1: Systems and methods for detecting hallucinations in machine learning models Date: published 6 March 2025; listed priority 31 August 2023. URL: https://patents.google.com/patent/US20250077940A1/en	Claims 1 and 5-6 cover verification prompts, conditional output and iterative improvement. They do not expressly recite the exact source-binding engine in the inspected claims. Generic verify/retry/gate workflows must not be claimed new.
P16 - ProvSQL: Probabilities / Knowledge Compilation documentation Date: undated living documentation, accessed 19 October 2026; not treated as dated historical proof for every documented feature. URLs: https://provsql.org/docs/user/probabilities.html ; https://provsql.org/docs/user/knowledge-compilation.html	Documents BID inputs, correlated provenance, exact evaluation and circuit/CNF/knowledge-compilation routes. Does not provide a validated implementation of our scientific-text front end. It establishes relevant existing software capabilities, not a temporal priority opinion.
P17 - Query-Guided Analysis and Mitigation of Data Verification Errors (Extended Version) Date: 9 March 2026. URLs: https://arxiv.org/abs/2603.08612 ; https://arxiv.org/html/2603.08612v1	Covers worst-case query-output error, risky inputs and targeted additional verification, including external verifiers. Does not establish the exact categorical header grammar used here. Error-aware targeted re-verification is not a safe novelty fallback.
P18 - US11113275B2: Verifying text summaries of relational data sets Date: grant publication 7 September 2021; listed priority 6 November 2017; related A1 published 1 July 2021. URL: https://patents.google.com/patent/US11113275B2/en	Describes mapping numerical text claims to relational queries, probabilistic inference, ambiguous translations, shared document context, user feedback and batched query evaluation. This is close to the combined use case, not merely one isolated component. It does not expressly establish our exact correlated-header all-world implementation in the inspected passages. Full family/claim construction remains outstanding.
P19 - Table-Text Alignment: Explaining Claim Verification Against Tables in Scientific Papers Date: 12 June 2025; revised 17 September 2025. URLs: https://arxiv.org/abs/2506.10486 ; https://arxiv.org/html/2506.10486v1	Covers scientific table/claim alignment and explanatory evidence. Does not establish the exact all-world state-merging and clarification protocol in the inspected material. Scientific-table evidence alignment is not a new application area.

Source / date / URL	Coverage and remaining limits
P20 - Thucy: An LLM-based Multi-Agent System for Claim Verification across Relational Databases Date: 2 December 2025; revised 5 January 2026. URL: https://arxiv.org/abs/2512.03278	Covers claim verification across relational tables/databases with concrete evidence. Abstract inspected; source code/experimental claims not reproduced. Does not establish our exact bounded source-choice representation. A database-backed claim verifier is not itself a novel combination.
P21 - Evidence-Grounded Verified Agentic Reasoning: A Path Toward Eliminating LLM Hallucination in Empirical Inference via Tool-Attested Kernel Proofs Date: 14 July 2026. URL: https://arxiv.org/abs/2607.12650	Covers evidence-grounded formal verification and tool-attested reasoning. Does not establish the exact header-choice engine in the abstract inspected this stage. "Proof-carrying" or "certified" output must not be claimed as a new general feature.
P22 - ToolSciVer: Multimodal Scientific Claim Verification with Visual Tool Augmented Reinforcement Learning Date: 17 July 2026. URL: https://arxiv.org/abs/2607.16131	Covers visual scientific verification with table/column focus, parsing and tool-efficiency objectives. Abstract inspected. Does not establish exact finite-world semantics in that material. Selectively inspecting source regions is not new by itself.

Legal context: L01, official Guidelines for Examination of Computer Related Inventions (CRIs), 2025, sections 3.5/4.5 and separate novelty/inventive-step discussion: https://ipindia.gov.in/storage/uploads/docs-operator/7fdd559b-1ff8-4a87-8c36-033e7356b39b.pdf . L02, official PIB release, 29 July 2025: https://www.pib.gov.in/PressReleasePage.aspx?PRID=2149719 . Technical measurements do not by themselves establish eligibility, novelty or inventive step under Section 3(k).

4. Summary and background of the invention (Address the gap / Novelty)

Scientific claims may combine valid numbers taken from incompatible experimental configurations. ScopeGuard represents each admissible metadata interpretation explicitly and evaluates whether all numerical conditions can hold in one configuration. It distinguishes supported, contradicted, insufficient and ambiguous outcomes under a declared finite-domain policy. Its new code compiles this policy once into a BDD and uses restrictions to evaluate possible header clarifications.

Stage 1 found the core mechanisms already covered by P01-P08/P16, and close claim-verification applications in P09-P15/P18-P22. P18 is particularly close to the combined use case: relational numerical claims, probabilistic ambiguity, context and feedback. Stage 2 rejected three candidate novelty claims. No affirmative claim of novelty is made in this completed form.

The remaining research gap is empirical: under which structured-input conditions does reusable compilation reduce clarification cost, and when do simpler methods win? E1-E3 in Question 8 establish a limited measured engineering effect, not a technical feature absent from P04/P06/P16. Neither scientific terminology nor exact implementation differences establish patent novelty. The proposed journal contribution is a candid comparative study, benchmark and failure analysis; publication acceptance is unproven.

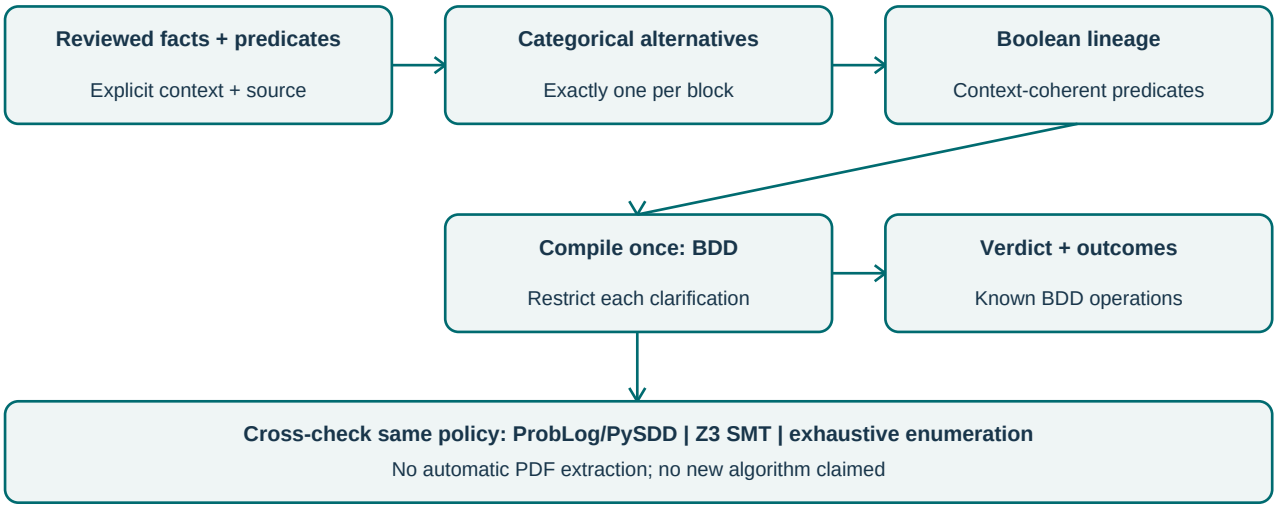
5. Objective(s) of Invention

Preserve experimental context across compound numerical conditions; expose verdict changes across admissible interpretations; reuse a compiled relation for conditional clarification queries; compare exact backends and document computation and memory costs; retain adverse cases and abstentions. These are research/engineering objectives, not statements that their underlying mechanisms are new.

6. Working principle of the invent (in brief)

Input consists of reviewed facts, numerical predicates, a finite set of contexts and mutually exclusive alternative fact groups. One Boolean selector represents each alternative; exactly-one constraints define admissible worlds. For each context and predicate the engine computes whether an observed value passes or fails. A world is supported when some single context satisfies every predicate. It is contradicted only when the declared domain is closed and every context has an observed violation. Otherwise it is insufficient. If different worlds yield different statuses, the overall answer is ambiguous. The compiled BDD shares these formulas; fixing a source-choice selector answers a proposed clarification without rebuilding the entire representation. A cheapest single clarification is suggested only if all its outcomes remove ambiguity; it is not a multi-step optimal policy.

7. Description of the invention in detail (Include drawing and or photograph as needed)



The pipeline is: manually reviewed source/table facts and predicates -> categorical source alternatives and context keys -> Boolean lineage with exactly-one constraints -> reusable BDD -> current verdict and conditional outcomes. Parallel adapters encode the same policy in ProbLog/PySDD and Z3, while exhaustive enumeration provides a small-instance oracle. These engines evaluate the same structured input; the diagram does not imply an automatic PDF extraction system.

Facts retain entity, metric, unit, exact numeric value, context and source locator. Input validation rejects conflicting values that can coexist at the same key. Numerical comparisons use exact decimal/rational values, not embedding thresholds. Choices are independent across blocks and mutually exclusive within each block; more general cross-block correlations are unsupported. Closed-domain contradiction applies only to the listed contexts. The paper fixtures deliberately limit that domain to selected rows.

Implementation: src/claimaudit/verification/compiled_scope.py contains reusable BDD compilation; src/claimaudit/evaluation/scope_backends.py contains the actual ProbLog, Z3, DP, rebuild and exhaustive adapters; src/claimaudit/evaluation/adversarial_benchmark.py isolates workers and records raw samples. tests/test_scope_backends.py tests equality across independently expressed engines. scripts/evaluate_local_scope_llm.py saves the optional local-model requests and responses. Exact paths and commands are preserved in CHANGESET.md and STAGE3_RESULTS.md.

Limits: no extraction recall evaluation, no guarantee that all plausible interpretations were captured, no calibrated probabilities, no arbitrary natural-language entailment, no automatic unit conversion, no optimum multi-step clarification plan, and no general polynomial complexity. BDD size may grow exponentially and depends on variable ordering. This extension does not replace the existing embedding/NLI project components; it evaluates an explicit structured subproblem.

8. Experimental validation results:

E0: 90 tests passed. E1: all six engines matched 36/36 reserved input labels; there were zero discrepancies in initial and single-choice conditional statuses against exhaustive verification where exhaustive completed. The 24 topology inputs use an existing exhaustive oracle; 12 paper-table fixtures were assistant-annotated from Attention Is All You Need, Table 3, and Deep Residual Learning for Image Recognition, Table 6. Independent human annotation remains absent.

E2 - reserved topology cost, exact integer nanoseconds and bytes:

Engine	Correct / completed / total	Sum of case median times (ns)	Max process RSS (bytes)
scope_dp	24 / 24 / 24	26714833	24608768
bdd_rebuild	24 / 24 / 24	104202459	38715392
compiled_reuse	24 / 24 / 24	24968876	39370752
z3	24 / 24 / 24	124363956	53886976
problog	24 / 24 / 24	2516006750	198213632
exhaustive	24 / 24 / 24	107458539	24264704

Times are sums of per-case medians from three isolated-process repetitions, including validation/compilation and all conditional queries; imports/startup are excluded. Compiled reuse reduced this aggregate relative to rebuilding, but this follows known BDD operations (P04/P06/P16). Its peak RSS is higher than rebuilding and original DP; no memory-saving claim survives.

E2 - paper fixtures, where repeated-query reuse has no advantage:

Engine	Correct / completed / total	Sum of case median times (ns)	Max process RSS (bytes)
scope_dp	12 / 12 / 12	433207	24150016
bdd_rebuild	12 / 12 / 12	889793	39141376
compiled_reuse	12 / 12 / 12	914541	38977536
z3	12 / 12 / 12	10831750	53641216
problog	12 / 12 / 12	93755334	194035712
exhaustive	12 / 12 / 12	147417	23904256

E3: Four adverse cases have 6, 10, 12 and 16 independent binary choices with distinct signatures. At 16, original DP was incomplete at its 5000-state cap and exhaustive was skipped above 4096 worlds. Compiled reuse, BDD rebuild, Z3 and ProbLog completed all four. Their adverse-16 median times were respectively 4464791, 60532125, 11274125 and 12451630959 ns. There were no final-run worker timeouts or errors. Favorable-16 also skipped exhaustive under the same cap; original DP took 113292 ns versus compiled reuse 1384083 ns. These adverse cases defeat DP merging, not BDD worst-case complexity. No result is extrapolated to larger instances.

E4: Local Ollama qwen3:4b (4B, Q4_K_M), one zero-shot prompt, seed 20261019, temperature 0, thinking enabled, 2048 generated-token cap, 8192 context cap. Correct 8/12; false supports 0; invalid/error outputs 4. Full requests, responses, timing/token metadata, model digest and protocol are saved. This is NOT a strong/frontier-model test; its results cannot be generalized to stronger models. No prompt tuning or retries were used. Invalid outputs count as incorrect under this fixed budget, not as demonstrated semantic errors or evidence splicing.

Paper claims are authored fixtures rather than claims of errors in the original papers. No false-support verdict occurred in this run; the targeted splicing error was not demonstrated. Invalid responses are a separate budget/format failure. Strong-LLM comparison, larger independent annotation and relevant-environment validation remain missing. Neither this experiment nor the sources support a patentability conclusion.

Reproducibility: seed 20261019; Python 3.12.11; dd 0.6.0; Z3 4.15.4.0; ProbLog 2.3.0; PySDD 1.0.6. Frozen input SHA-256: 02e1ded15415be7e90f95bc28da336cf6f7567e55fd69526bef3f35da60528c4. Raw solver samples, limits, hashes and all local-model responses are saved with the code. STAGE3_RESULTS.md contains commands and interpretation; no rounding or fabricated results were used.

9. What aspect(s) of the invention need(s) protection?

No presently demonstrated technical aspect is recommended as an established patent-novel claim. Candidates for professional assessment only: (i) the exact source-interpretation policy and its implementation boundary, assessed against P01-P08/P12/P18 and E1; (ii) the particular compile/restrict integration, assessed against P04/P06/P16 and E2-E3. Both failed this preliminary novelty screening. They are listed to make the assessment boundary explicit, not as proposed patent claims. A professional may assess claim scope, ownership and any additional unpublished technical material; none is presumed here.

Must NOT be claimed as new: BID/x-tuples, possible-world/certain-answer semantics, equivalent-state merging, BDD restriction, generic query-guided or budgeted cleaning, claim decomposition, numerical provenance, ambiguity reporting, general solver-assisted verification, generic explanation/gating, or the evidence-splicing failure mode. The ScopeGuard name and a measured speed difference do not establish invention. Copyright/licensing and dataset documentation are separate issues and do not cure the absence of patent novelty.

Open risks: some exact publication dates are unverified; certain sources were inspected only at abstract level; failed URLs were excluded rather than fabricated. Patent search used Google Patents and did not exhaust InPASS, PATENTSCOPE, Espacenet, families, claims, prosecution histories, non-English documents or unpublished applications. Independent annotation, strong-model evidence, extraction robustness and BDD-hard cases remain missing. Consult the institutional IP cell/professional about disclosure timing before any public manuscript, repository or demonstration; prior public disclosures by others or the user have not been audited. No public release or patent filing was performed in this task.

10. What is Technology readiness level of your invention? (Tick the appropriate TRL)

☐	☐	☑	☐	☐	☐	☐	☐	☐
TRL 1	TRL 2	**TRL 3**	TRL 4	TRL 5	TRL 6	TRL 7	TRL 8	TRL 9

TRL 3 - Experimental proof of concept. Selected because executable algorithms, cross-engine checks and limited controlled experiments exist. No real-environment validation is available. TRL 4-9 are not claimed; the existence of a demonstration UI does not establish operational readiness.